

TIMER AND INTERRUPT MECHANISM

MS51FB9AE Microcontroller Simulator

1. Timer Mechanism

A **timer** is a peripheral used to measure time or count clock cycles/events. In the MS51FB9AE Microcontroller Simulator, the timer is part of the **GPIO/Timer module** and is updated by the **Scheduler** during simulation.

Working of Timer

- The timer starts with an initial count value.
- The Scheduler updates the timer during each simulation cycle.
- The timer counter increments or decrements depending on its configuration.
- When the predefined limit is reached, a timer event occurs.
- The event can generate an interrupt.
- The CPU then handles the interrupt before continuing normal execution.

Example:

Timer Limit = 5

Cycle 1 → 1

Cycle 2 → 2

Cycle 3 → 3

Cycle 4 → 4

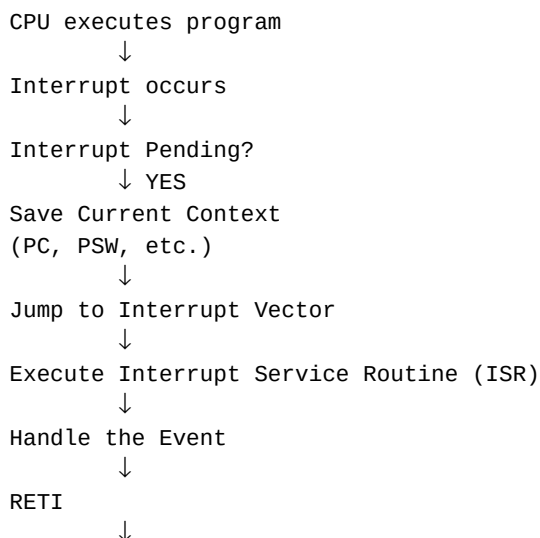
Cycle 5 → 5 → Timer Event → Interrupt

Timers can therefore be used to simulate delays, periodic events, counters, and time-based operations.

2. Interrupt Mechanism

An **interrupt** is a signal that temporarily stops the normal execution of a program so that the CPU can handle an important event. In the simulator, interrupts may originate from peripherals such as the **Timer** or **GPIO**.

Interrupt Working



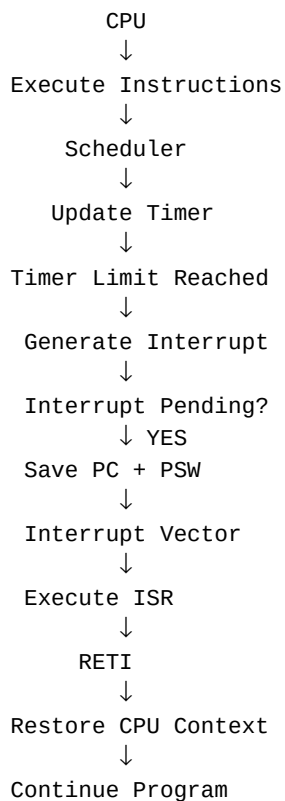
Restore Previous Context
↓
Continue Program

Important Components

Component	Function
Interrupt Source	Generates the interrupt, such as a timer.
Interrupt Controller	Detects and manages interrupt requests.
Interrupt Enable	Determines whether an interrupt is allowed.
Interrupt Vector	Provides the address of the corresponding ISR.
Stack	Temporarily stores CPU information such as PC and PSW.
ISR	Handles the interrupt event.
RETI	Returns from the interrupt to normal program execution.

3. Timer and Interrupt Together

The timer and interrupt mechanisms work closely together. For example, when a timer reaches its predefined limit, it can generate an interrupt. The CPU saves its current state, executes the appropriate ISR, and then returns to the interrupted program.



Conclusion

The **timer** provides time-based or counting functionality, while the **interrupt mechanism** allows the CPU to respond to important events. Together, they make the MS51FB9AE simulator capable of demonstrating realistic microcontroller timing, event handling, and CPU control flow.